

Limit Test

Definition, Limit test on Ash Value, Limit test on Moisture Content

Contents

- **Definition and importance**
- **Limits on ash values**
- **Limits on moisture content**

Definition and Importance

- **Limit tests are quantitative or semi-quantitative tests designed to identify and control small quantities of impurities which is likely to present in the substance.**
- **The test involves comparing the sample against a known standard or limit to determine if the concentration of the substance being tested falls within an acceptable range. If the concentration exceeds the limit, it indicates a potential quality issue or impurity in the sample.**

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- It is widely used to ensure the safety, quality and efficacy of drug products by identifying impurities.
- There are different limit tests including tests for chloride, iron, sulphates, arsenic and those for heavy metals.

Limits test on ash values

- **Ash Value is defined as the amount of inorganic residue that is left over when a sample is burned completely. It represents the mineral content in a sample.**
- **Ash value limit test determines the amount of residual inorganic matter left after ignition of a substance.**
- **It's widely used to check the purity and quality of products like tablets, capsules, and cosmetics.**
- **Ash value is determined by weighing the residue left after igniting the sample. The result is expressed as a percentage by weight.**

Determining Ash Value

- There are four methods
 1. Direct Ashing method
 2. Wet Ashing method
 3. Loss on ignition method
 4. Microwave Ashing method

1. Direct Ashing Method

- **This approach involves burning a sample of known weight at a high temperature in a muffle furnace.**
- **The sample is burned until all of the organic material has been removed, leaving just ash or inorganic residue. After weighing the residual ash, the ash value is determined as a percentage of the initial sample weight.**

2. Wet Ashing Method

- This technique involves digesting the sample using strong acids like nitric or sulphuric acid.**
- The acid digestion process breaks down organic matter and converts it into soluble compounds, leaving behind the inorganic ash.**
- The resulting solution is filtered, and the residue is washed, dried, and weighed to determine the ash content. The ash value is calculated as a percentage of the original sample weight.**

3. Loss on ignition method

- This method is usually applied to drugs which contain volatile or combustible substances.
- A known weight of the sample is heated at a specific temperature (typically between 500°C and 800°C) in a furnace or oven.
- The sample is heated until all volatile components are burned off, leaving behind the non-volatile ash. The remaining ash is weighed and the ash value is calculated as a percentage of the original sample weight

4. Microwave Radiation Method

- In this method, the sample is placed in a microwave digestion vessel along with suitable reagents, such as strong acids or oxidizing agents.**
- The vessel is then subjected to microwave radiation, which rapidly heats the sample and facilitates the combustion of organic matter.**
- After cooling, the residue is filtered, washed, dried, and weighed to determine the ash content.**

Advantages of Ash values

- **It is useful for detecting low grade products, exhausted products, excess of sandy and earthy matter with drug.**
- **Simple and inexpensive**
- **Indication of inorganic contents**
- **Detection of adulteration**

Disadvantages

- Not suitable for all drugs
- Limited specificity
- Not always reliable
- Not a definitive purity indicator

Procedure

1. Weighing the sample and crucible
2. Incineration
3. Cooling
4. Weighing the crucible containing the Ash
5. Calculate the Ash value by subtracting the weight of the empty crucible from the weight of crucible containing the ash.

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$$\% \text{ Ash value} = \frac{\text{Weight of ash}}{\text{Weight of sample}} \times 100\%$$

Types of Ash values

- ☐ Total Ash value
- ☐ Water insoluble Ash value
- ☐ Acid insoluble Ash value
- ☐ sulphate Ash value

Limits test on moisture content

- **Moisture content is the amount of water content in a substance. It is typically expressed as a percentage of the weight of the sample.**
- **Presence of moisture markedly affects chemical stability, texture and shelf life of pharmaceutical products. Therefore, it is essential to limit and quantify the amount of water present in different pharmaceuticals.**
- **It is one of the key aspects in quality control.**

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- **Moisture accelerates**
 - the hydrolysis of the drug
 - reaction with other agents and excipients
 - Reducing the stability of the drug.
- **Limiting the moisture content of the drug is particularly important for solid dosage forms due to the reason that solids have a great tendency to absorb moisture.**

Methods of moisture content determination

1. loss on drying method
2. Karl-Fischer titration method
3. Azeotropic distillation method

- Other methods also can be used but the above are widely used(specially the above 2).

1. Loss on drying method

- A known weight of the sample is placed in an oven at a specific temperature (typically between 105°C and 110°C) for a defined period of time.
- The sample is dried until a constant weight is achieved, indicating that all moisture has been removed. The moisture content is calculated as the percentage loss in weight of the sample.

2. Karl-Fischer titration method

- It is a specific method used for the determination of water content in samples with low moisture levels.
- Named after German chemist Karl Fischer, who developed it in 1935, this method is highly valued in industries.
- It involves the reaction of water with a Karl Fischer reagent, which contains iodine, sulfur dioxide, pyridine, or imidazole, and a suitable solvent.
- The amount of reagent required to reach the endpoint is proportional to the water content in the sample.



In this reaction:

- Water (H_2O) is the target substance.
- Iodine (I_2) reacts with water.
- Sulfur dioxide (SO_2) and the organic base (RN) act as catalysts, enabling a complete reaction.
- The end products, hydriodic acid (HI) and organic sulfate ($\text{RN-SO}_3\text{R}$), are produced once all water has reacted.

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- There are two main types of Karl Fischer titrations: volumetric and coulometric.

❖ Volumetric

- Iodine is added to the sample through a burette.
- The amount of water is calculated based on the volume of KF reagent consumed to reach the endpoint.
- This approach is suitable for samples with water content typically ranging from about 0.1% to 100%.

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❖ Coulometric

- This approach is ideal for low water content samples, typically in the range of 10 µg to 10 mg of water (0.001% to 5%).
- Instead of adding iodine directly, iodine is generated electrochemically in the titration cell by applying a current.
- The amount of iodine generated is proportional to the electric current passed, allowing for precise calculation of water content.

3. Azeotropic Distillation Method

- **It is a method commonly used in the determination of moisture content in samples that are difficult to dry or contain volatile components.**
- **A suitable solvent is added to the distillation apparatus containing the sample. The solvent should form an azeotropic mixture with water, meaning that it has a boiling point close to that of water and forms a constant-boiling mixture.**
- **Common solvents used for azeotropic distillation include toluene, benzene, or cyclohexane.**

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- The distillation apparatus is connected to a condenser, and the mixture is heated. As the mixture is heated, the solvent and water form an azeotropic vapor mixture that distills over. The water vapor carries the moisture from the sample into the condenser.
- The water-containing vapor mixture is condensed in the condenser, converting it back into liquid form. The condensed liquid is collected in a separate receiver or trap.
- The collected liquid is analyzed to determine the moisture content. It is calculated as the percentage of the weight of the moisture collected in the receiver to the initial weight of the sample.

4. Other methods

- **Microwave Moisture Analysis**
- **Infrared (IR) Moisture Balance**

References

- *" Pharmaceutical analysis "* by Beckett and stenlake
- *" analytical chemistry for pharmaceutical science" by Skoog ,Holler, and Crouch.*
- *Researches from different scholars.*

Thank You